

Q1. Work in the charging phase

The external torque must overcome the restoring torque $-\theta$ at every instant, so the work it does is

$$W = \int_0^{\theta_0} k\theta \, d\theta = \boxed{\frac{1}{2}k\theta_0^2}.$$

Q2. Rotational energy and angular speed

All the work done in charging is stored in the spring and, at release, converted to the bolt's rotational kinetic energy. The arm is massless, so the moment of inertia about the pivot is $I = mL^2$:

$$\frac{1}{2}k\theta_0^2 = \frac{1}{2}I\omega^2 = \frac{1}{2}mL^2\omega^2.$$

i. $\omega = \theta_0 \sqrt{\frac{k}{mL^2}} = \boxed{\frac{\theta_0}{L} \sqrt{\frac{k}{m}}}.$

ii. $v = \omega L = \boxed{\theta_0 \sqrt{\frac{k}{m}}}.$

iii. $a_c = \omega^2 L = \boxed{\frac{\theta_0^2 k}{mL}}.$

Q3. Angular momentum: conserved or not?

Wrong. The angular-momentum theorem states $\frac{dL}{dt} = \tau_{\text{net}}$, where τ_{net} is the net external torque about the pivot. During charging the torsional spring exerts a nonzero torque $-\theta$ on the arm, and this torque is not balanced by anything — it is precisely the external torque being worked against. Since $\tau_{\text{net}} \neq 0$, the angular momentum is *not* conserved. (After release, gravity also exerts a moment about the original pivot on the freely flying bolt, further changing its angular momentum.)

Q4. Projectile motion and mechanical energy

- i. After release the only force doing work is gravity, so mechanical energy is conserved:

$$\frac{1}{2}mv_f^2 = \frac{1}{2}mv^2 + mgH \implies \boxed{v_f = \sqrt{v^2 + 2gH}}.$$

Substituting $v = \theta_0 \sqrt{k/m}$ gives $v_f = \sqrt{\frac{\theta_0^2 k}{m} + 2gH}.$

- ii. For small θ_0 , the horizontal range *increases* as θ_0 increases. Physically, the horizontal velocity $v \cos \theta_0$ barely changes (it is $\approx v$), while the upward velocity $v \sin \theta_0$ grows roughly linearly with θ_0 ; a larger upward velocity means a longer time of flight, which dominates, so the bolt travels farther. (The exact range from height H grows as $\frac{v^2}{g}\theta_0 + v\sqrt{2H/g}$ for small θ_0 .)

Q5. Model extension: critical equilibrium

With the bolt at the tip, gravity exerts a torque about the pivot equal to $-mgL \cos \theta$ (the lever arm is the horizontal distance $L \cos \theta$ from the pivot to the bolt). The spring exerts $-k\theta$, so static equilibrium requires

$$-k\theta - mgL \cos \theta = 0 \implies \boxed{k\theta + mgL \cos \theta = 0}.$$

For $\theta > 0$ both torques point the same way (back toward horizontal), so there is no equilibrium above horizontal. The balance is possible only *below* horizontal, where the spring pulls upward and gravity pulls downward; there the equilibrium angle solves the transcendental equation

$$\boxed{\theta = -\frac{mgL}{k} \cos \theta}.$$